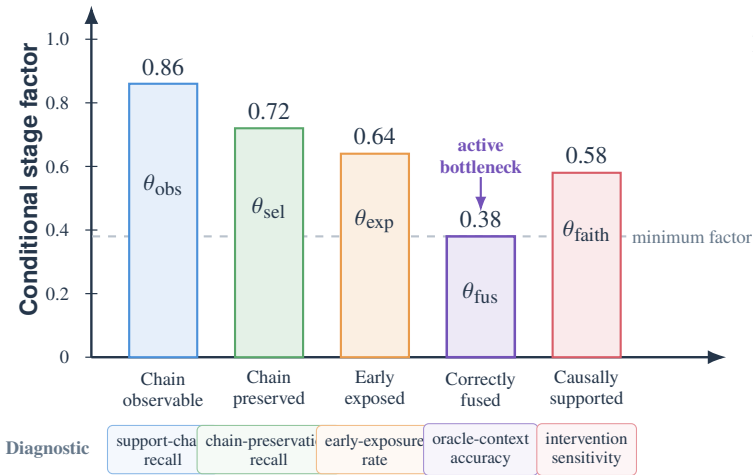




Illustrative values (not empirical results)

Conditional factors

$$\theta = (0.86, 0.72, 0.64, 0.38, 0.58)$$

Conditional bottleneck bound

$$P(Y_g) \leq \min_i \theta_i = 0.38$$

Grounded-success product

$$P(Y_g) = \prod_i \theta_i \approx 0.087$$

Coordinate sensitivity at the bottleneck

$$\partial P(Y_g) / \partial \theta_{\text{fus}} \approx 0.230$$

Active bottleneck: fusion survival

The bound is diagnostic: improving one component does not hold the others fixed in a real system.